Ex. No: 13
24/9/26

Implement your own ping program

Aim:-
To implement a simple ping program in JAVA to check whether a host is reachable and calculate the response time.

Algorithm:-
1) Read the hostname or IP address from the user
2) Get the IP address using Inet Address
3) send a ping request using is Reachable()
4) Record the start and end time.
5) calculate the response time.
6) Display whether the host is reachable.

Program:-

```
import JAVA.net.InetAddress;
import java.util.Scanner;
Public class myping {
    Public static void main (string [] args) {
        Scanner sc = new Scanner (system.in);
        system.out.print ("Enter host name or IP address:");
        string host = sc.readline();
    try {
        Inet Address address = Inet Address.getByName (host);
        system.out.println ("pinging " + address.get host Address())
    for (int i=1; i<=1; i++) {
```

Output:

Enter host name or IP address: google.com

pinging 142.250.183.14

Reply from 142.250.183.14: time = 32 ms

Reply from 142.250.183.14: time = 29 ms

Reply from 142.250.183.14: time = 31 ms

Reply from 142.250.183.14: time = 30 ms

```
long start = system.currentTimeMillis();
boolean reachable = address.isReachable(3000);
long end = system.currentTimeMillis();
long time = end - start;
if (reachable){
    system.out.println("Reply from" + address.
        getHostAddress() + ": time=" + time
        + "ms");
} else {
    system.out.println("Request timed out.");
}
Thread.sleep(1000);
}
catch (Exception e) {
    system.out.println("Error:" + e.getMessage());
}
sc.close();
}
}
```

Result :-

Thus, a simple Java program was successfully implemented in JAVA to check host reachability and measure response time.